Goldman Sachs Research

Economics Research
29 May 2026 | 12:25PM EDT

THE WORLD CUP AND ECONOMICS

World Cup 2026: Predictions, Probabilities, and Paths to Victory

- We present a model for predicting the outcome of the 2026 World Cup in the US, Canada and Mexico from June 11 to July 19. It is similar to our model for prior World Cups but extended along several dimensions.
- Historical performance data for each team—most importantly the Elo rating system originally devised to rank chess players—is the main determinant of the number of goals scored. But even for a given Elo rating, we show that scoring talent and team momentum matter, as well as several mentality and geographical factors.
- We then simulate a set of probabilities that a particular team will reach a particular round, up to winning the World Cup. We also provide a modal forecast for how the tournament will unfold. After each day of play, we will re-run the model using updated performance data in order to generate new probabilities and a new modal forecast.
- The model says that Spain has a 26% probability of winning the trophy, followed by France at 19%, Argentina at 14%, Brazil at 8% and England at 5%. Spain is predicted to win because it has the highest Elo ranking, supported by scoring talent and good momentum into the competition. Argentina is penalised by the "winner's slump", i.e. the statistical underperformance of reigning champions in the following World Cup; France suffers from likely facing top-ranked Spain in the semifinals; and England underperforms its Elo rating given historical tournament disappointment, geographical headwinds (likely facing Mexico in high-altitude Mexico City), and a slightly unlucky draw.
- How much faith should we have in these predictions? Our projections are not far from bookmakers' odds—except for a lower winning probability for England—and our model would have performed quite well at previous World Cups (for example, when measured with the goal difference). That said, the model's statistical power remains limited—not a surprise given football's inherent unpredictability.

Jan Hatzius
+1(212)902-0394 | jan.hatzius@gs.com
Goldman Sachs & Co. LLC

Giovanni Pierdomenico
+44(20)7051-6807 |
giovanni.pierdomenico@gs.com
Goldman Sachs International

Sven Jari Stehn
+44(20)7774-8061 | jari.stehn@gs.com
Goldman Sachs International

World Cup 2026: Predictions, Probabilities, and Paths to Victory

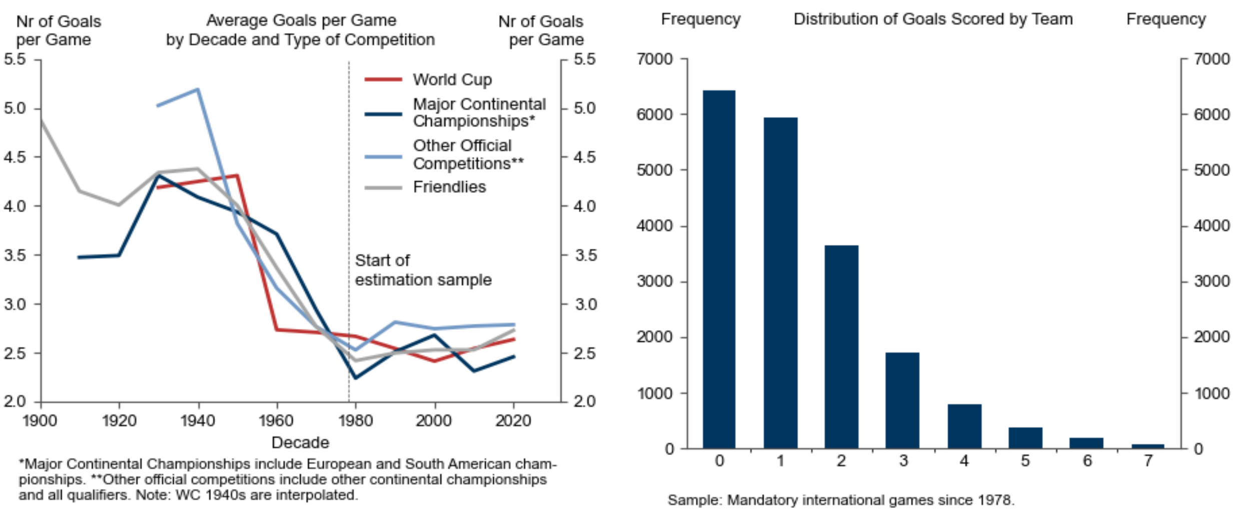
“Football is the most important of the least important things in life” - Arrigo Sacchi, Italian former football manager

With two weeks to go, we introduce our statistical model for predicting the outcome of the 2026 World Cup. We proceed as follows.

Our Statistical Model to Predict the Outcome of the World Cup

First, we estimate a regression model to predict the number of goals scored by a particular team (“team i”) against a particular opponent (“team j”) using the entire history of mandatory international matches since 1978 (a total of just under 20,000 matches). Following the literature on predicting football matches, we assume that the number of goals scored by team i is described by a so-called “Poisson” distribution (Exhibit 1).

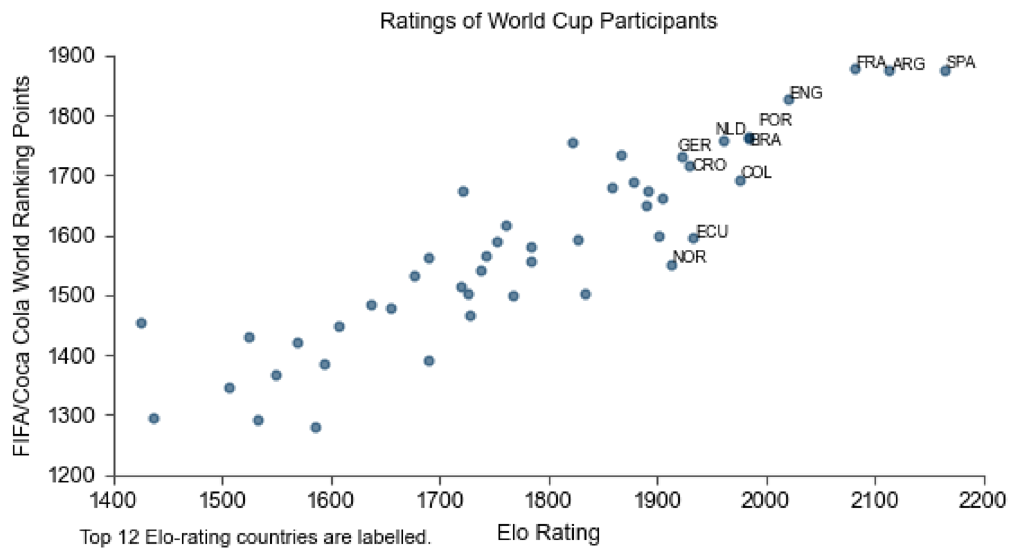
Exhibit 1: The Number of Goals Score by a Football Team Broadly Follows a Poisson Distribution



Source: eloratings.net, Goldman Sachs Global Investment Research

The main determinant of the number of goals scored is the difference in team performance as reflected in Elo ratings prior to the match. The Elo system was originally devised to rank chess players. It is a composite measure of national football team success that evolves depending on a team’s results and the strength of its opponents. The Elo system ranks Spain first, followed by Argentina and France, and thus differs slightly from the FIFA men’s world rankings (Exhibit 2).

Exhibit 2: Elo vs FIFA/Coca-Cola Rankings



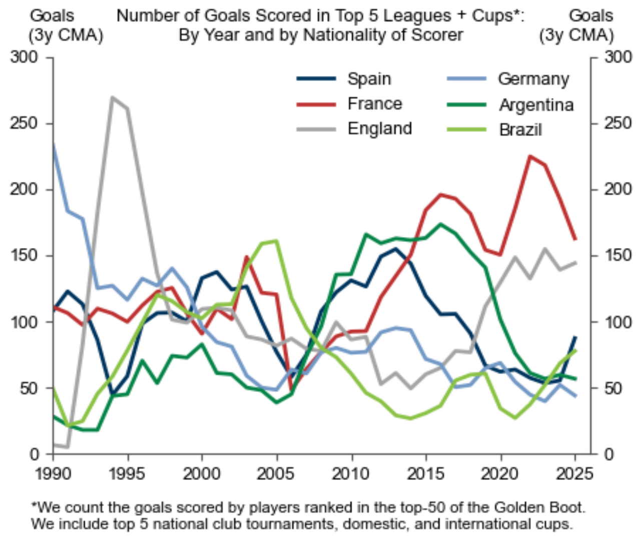
Source: eloratings.net, FIFA, Goldman Sachs Global Investment Research

We then expand the model with several other explanatory variables:

- **Scoring talent**: we gather data on top goal scorers in national leagues and international competitions for clubs and find that these matter for the number of goals scored by the team in a World Cup[1].
- **Momentum**: we measure team momentum by including the number of goals scored by team i in the last 10 competitive matches, and the number of goals conceded by team j in the last 5 competitive matches.
- **Mentality**: we find that reigning world champions typically underperform (while teams at their first World Cup usually outperform) and that it is more difficult to score against European teams (all else equal). Furthermore, major footballing nations generally get a World Cup boost, with the notable exception of England.
- **Geography**: we confirm a familiar "home effect" for games that are played at home. We also find that the number of goals scored is reduced: by the distance between the country's capital and location of the game; when a low-altitude country plays in a high-altitude stadium (which is relevant for games played in Mexico in the upcoming World Cup); and by the absolute temperature difference between the average temperature of a country and the temperature of the venue in the month of the game.

[1] We consider a counter-variable for the number of players in each national team ranked in the top 50 scorers in the top European club competitions, capped at 4 per team as it is rare to see more than 4 strikers on the pitch at the same time. Adding top scorers in the Copa Libertadores does not add to the fit.

Exhibit 3: Goal-Scoring Potential Matters; Some Signs of a "Winner's Slump" Effect After Winning the World Cup

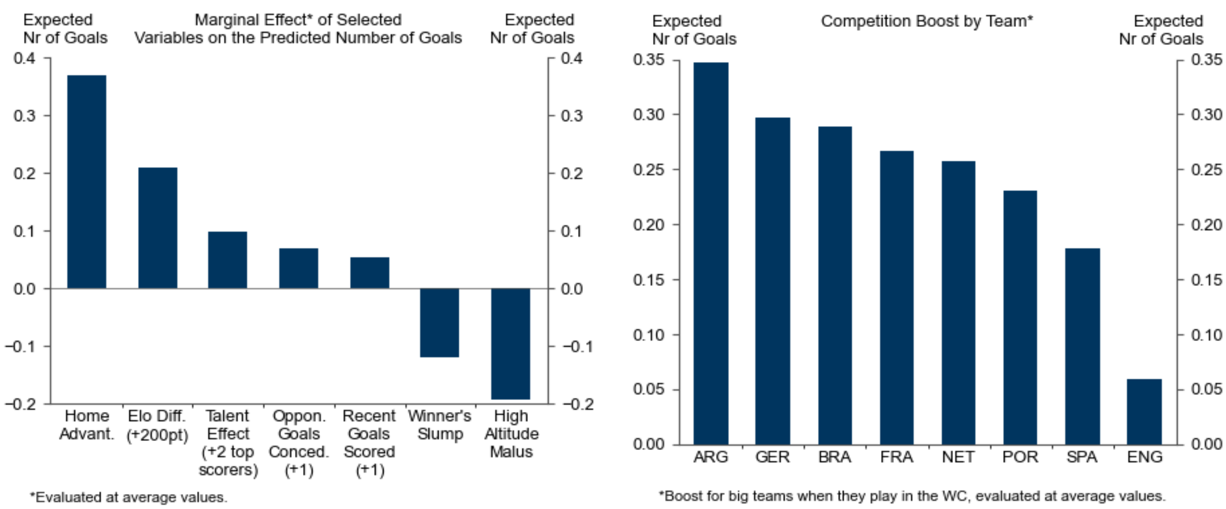


World Cup	Winner	Outcome in the Following World Cup
1978	Argentina	2nd Round
1982	Italy	Round of 16
1986	Argentina	2nd Place
1990	Germany	Quarter Finals
1994	Brazil	2nd Place
1998	France	Group Stage
2002	Brazil	Quarter Finals
2006	Italy	Group Stage
2010	Spain	Group Stage
2014	Germany	Group Stage
2018	France	2nd Place
2022	Argentina	TBC

Source: Transfermarkt, FIFA, Goldman Sachs Global Investment Research

Exhibit 4 summarises the marginal effect of some of the key variables in our model on the predicted number of goals scored in a World Cup game.

Exhibit 4: What Matters in Our Model



Source: Goldman Sachs Global Investment Research

Second, we apply this regression-based model to the upcoming World Cup. This entails simulating the group-stage games and the resulting bracket of the knock-out stage up to the final held on July 19 in New York. Our model updates Elo ratings and momentum variables recursively over the course of the tournament, according to model-based or, later, actual scores. We apply official FIFA rules for the tie-breakers in the group stage and account for all 495 possible pairings in the round of 32, which will depend on which third-placed teams qualify for the knock-out stage. We use the rounded prediction of the goals scored to determine the outcome of each match during the group stage and the unrounded prediction to pick the winner in the knockout stage[2].

The model thus generates the single most likely prediction for each of the 104 games in the World Cup up to the final. We can also leverage the Poisson distribution underlying our model to run a Monte Carlo simulation with 50,000 draws, which yields a set of probabilities that a particular team wins the tournament or reaches a particular stage of the tournament. In each of the simulations, the outcome of a game is given by a random draw (as opposed to the most likely) from the team- and game-specific Poisson distribution of goals scored[3].

A Summary of the Predictions

Exhibit 5 and Exhibit 6 show the most likely development of the tournament according to our model.

We expect major teams and all three hosts to advance from the expanded 48-team group stage to the round of 32. Groups D and G are expected to be the tightest ones, with our model unable to clearly differentiate (on a rounded basis) between the teams on the basis of the statistical factors considered. Exhibit 12 in the Appendix shows the predicted standings at the group stage.

We expect the round of 32 to offer a few interesting games, including the US vs. Iran and a *clásico* Argentina vs. Uruguay. That said, the first "big matches" should be featured in the round of 16, with Germany projected to be eliminated by France. England vs. Brazil and Argentina vs. Portugal (which could be the last match-up between Lionel Messi and Cristiano Ronaldo) should be the most exciting quarter-finals, with the *Seleção and the Selección* advancing to the next stage. The Netherlands and the tournament surprise Türkiye are also expected to end their run in the quarter-finals.

The model projects semi-finals of France vs. Spain, and Brazil vs. Argentina, with la *Roja* and la *Albiceleste* (reigning continental champions respectively in Europe and South America) facing off in a *Finalissima*. We expect Spain to win its second World Cup on July 19, with Messi (who turns 39 during the tournament) passing on the torch to Lamine Yamal (who will celebrate his 19th birthday just a few days before the New York final). Our prediction aligns with the historical pattern that the World Cup almost always comes back to Europe after having been won by a South American team (with the only exception being the consecutive Brazilian wins in 1958 and 1962).

[2] We implicitly assume that our unrounded prediction is a good proxy for the outcome of a game in extra time or at penalties after a draw in the 90 minutes of regular time.

[3] In the knock-out stage, we avoid draws by "burning" Monte Carlo draws that involve the same number of goals by the two teams. This mechanically boosts the chances of the team with the slightly higher mean predicted number of goals.

Exhibit 5: Most Likely Predicted Group Stage Results

FIFA World Cup 2026 - Group Stage: Prediction as of 29 May 2026

Played (final score) | Predicted score

Group A *Mexico | South Korea | Czechia | South Africa*

MD	Date	Team 1	Score	Team 2
1	11-Jun	Mexico	2-0	South Africa
1	11-Jun	South Korea	1-1	Czechia
2	18-Jun	Czechia	1-1	South Africa
2	18-Jun	Mexico	2-1	South Korea
3	24-Jun	Czechia	1-2	Mexico
3	24-Jun	South Africa	1-1	South Korea

Group B *Canada | Qatar | Switzerland | Bosnia and H.*

MD	Date	Team 1	Score	Team 2
1	12-Jun	Canada	2-1	Bosnia and H.
1	13-Jun	Qatar	0-2	Switzerland
2	18-Jun	Switzerland	1-1	Bosnia and H.
2	18-Jun	Canada	2-1	Qatar
3	24-Jun	Switzerland	1-1	Canada
3	24-Jun	Bosnia and H.	1-1	Qatar

Group C *Brazil | Haiti | Scotland | Morocco*

MD	Date	Team 1	Score	Team 2
1	13-Jun	Brazil	2-1	Morocco
1	13-Jun	Haiti	1-1	Scotland
2	19-Jun	Scotland	1-1	Morocco
2	19-Jun	Brazil	2-1	Haiti
3	24-Jun	Scotland	1-2	Brazil
3	24-Jun	Morocco	1-1	Haiti

Group D *USA | Australia | Turkiye | Paraguay*

MD	Date	Team 1	Score	Team 2
1	12-Jun	USA	1-1	Paraguay
1	13-Jun	Australia	1-1	Turkiye
2	19-Jun	USA	1-1	Australia
2	19-Jun	Turkiye	1-1	Paraguay
3	25-Jun	Turkiye	1-1	USA
3	25-Jun	Paraguay	1-1	Australia

Group E *Germany | Ivory Coast | Ecuador | Curacao*

MD	Date	Team 1	Score	Team 2
1	14-Jun	Germany	2-1	Curacao
1	14-Jun	Ivory Coast	1-1	Ecuador
2	20-Jun	Germany	2-1	Ivory Coast
2	20-Jun	Ecuador	2-1	Curacao
3	25-Jun	Curacao	1-1	Ivory Coast
3	25-Jun	Ecuador	1-2	Germany

Group F *Netherlands | Sweden | Tunisia | Japan*

MD	Date	Team 1	Score	Team 2
1	14-Jun	Netherlands	1-1	Japan
1	14-Jun	Sweden	1-1	Tunisia
2	20-Jun	Netherlands	2-1	Sweden
2	20-Jun	Tunisia	1-1	Japan
3	25-Jun	Japan	1-1	Sweden
3	25-Jun	Tunisia	1-2	Netherlands

Group G *Belgium | Iran | New Zealand | Egypt*

MD	Date	Team 1	Score	Team 2
1	15-Jun	Belgium	1-1	Egypt
1	15-Jun	Iran	1-1	New Zealand
2	21-Jun	Belgium	1-1	Iran
2	21-Jun	New Zealand	1-1	Egypt
3	26-Jun	Egypt	1-1	Iran
3	26-Jun	New Zealand	1-1	Belgium

Group H *Spain | Saudi Arabia | Uruguay | Cape Verde*

MD	Date	Team 1	Score	Team 2
1	15-Jun	Spain	3-0	Cape Verde
1	15-Jun	Saudi Arabia	1-1	Uruguay
2	21-Jun	Spain	3-0	Saudi Arabia
2	21-Jun	Uruguay	1-1	Cape Verde
3	26-Jun	Uruguay	0-2	Spain
3	26-Jun	Cape Verde	1-1	Saudi Arabia

Group I *France | Iraq | Norway | Senegal*

MD	Date	Team 1	Score	Team 2
1	16-Jun	France	2-1	Senegal
1	16-Jun	Iraq	1-2	Norway
2	22-Jun	France	3-0	Iraq
2	22-Jun	Norway	1-1	Senegal
3	26-Jun	Norway	1-2	France
3	26-Jun	Senegal	2-1	Iraq

Group J *Argentina | Austria | Jordan | Algeria*

MD	Date	Team 1	Score	Team 2
1	16-Jun	Argentina	2-1	Algeria
1	16-Jun	Austria	1-1	Jordan
2	22-Jun	Argentina	2-1	Austria
2	22-Jun	Jordan	1-1	Algeria
3	27-Jun	Algeria	1-1	Austria
3	27-Jun	Jordan	1-2	Argentina

Group K *Portugal | Uzbekistan | Colombia | DR Congo*

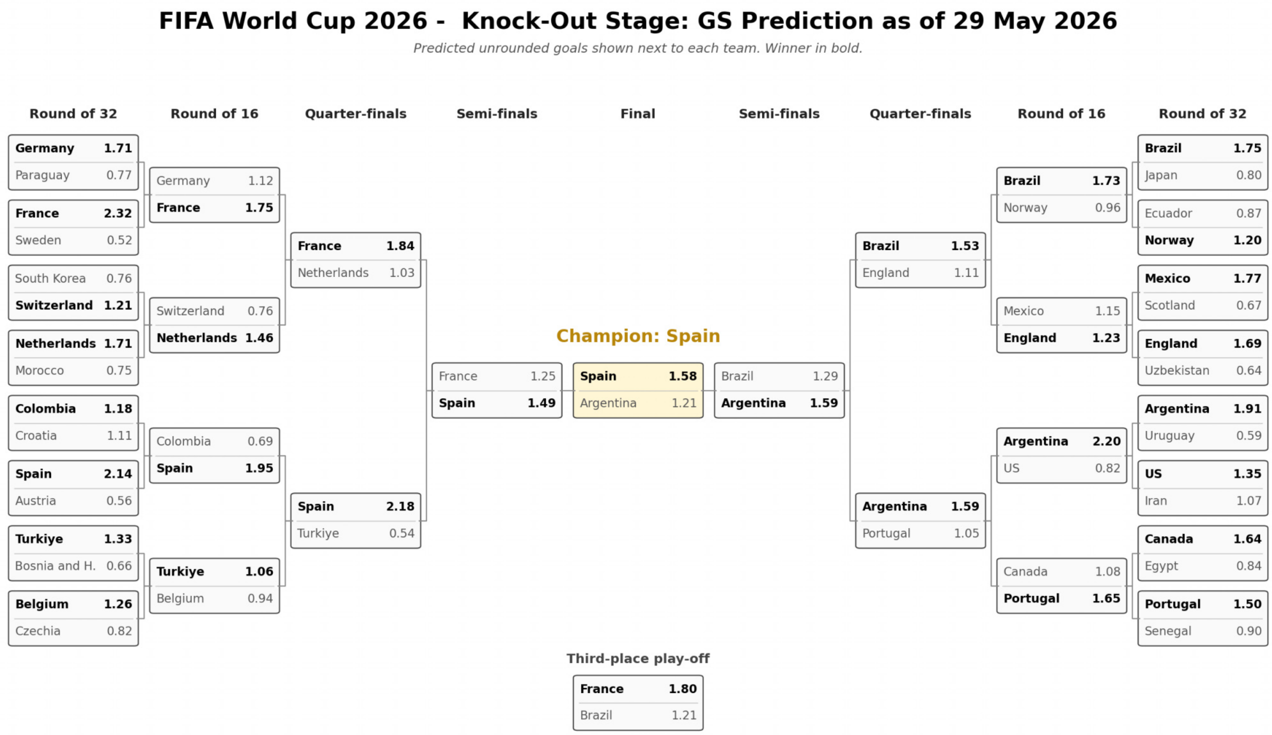
MD	Date	Team 1	Score	Team 2
1	17-Jun	Portugal	2-1	DR Congo
1	17-Jun	Uzbekistan	1-1	Colombia
2	23-Jun	Portugal	2-1	Uzbekistan
2	23-Jun	Colombia	2-1	DR Congo
3	27-Jun	Colombia	1-1	Portugal
3	27-Jun	DR Congo	1-1	Uzbekistan

Group L *England | Ghana | Panama | Croatia*

MD	Date	Team 1	Score	Team 2
1	17-Jun	England	1-1	Croatia
1	17-Jun	Ghana	1-1	Panama
2	23-Jun	England	2-1	Ghana
2	23-Jun	Panama	1-2	Croatia
3	27-Jun	Panama	1-2	England
3	27-Jun	Croatia	2-1	Ghana

Source: Goldman Sachs Global Investment Research

Exhibit 6: The Road to the Final



Source: Goldman Sachs Global Investment Research

The probabilities resulting from our Monte Carlo simulation are shown in Exhibit 7. The model says that Spain has a 26% probability of winning the trophy, followed by France at 19%, Argentina at 14%, Brazil at 8% and England and the Netherlands around 5%.

Exhibit 7: Probabilities of Advancement in the 2026 World Cup

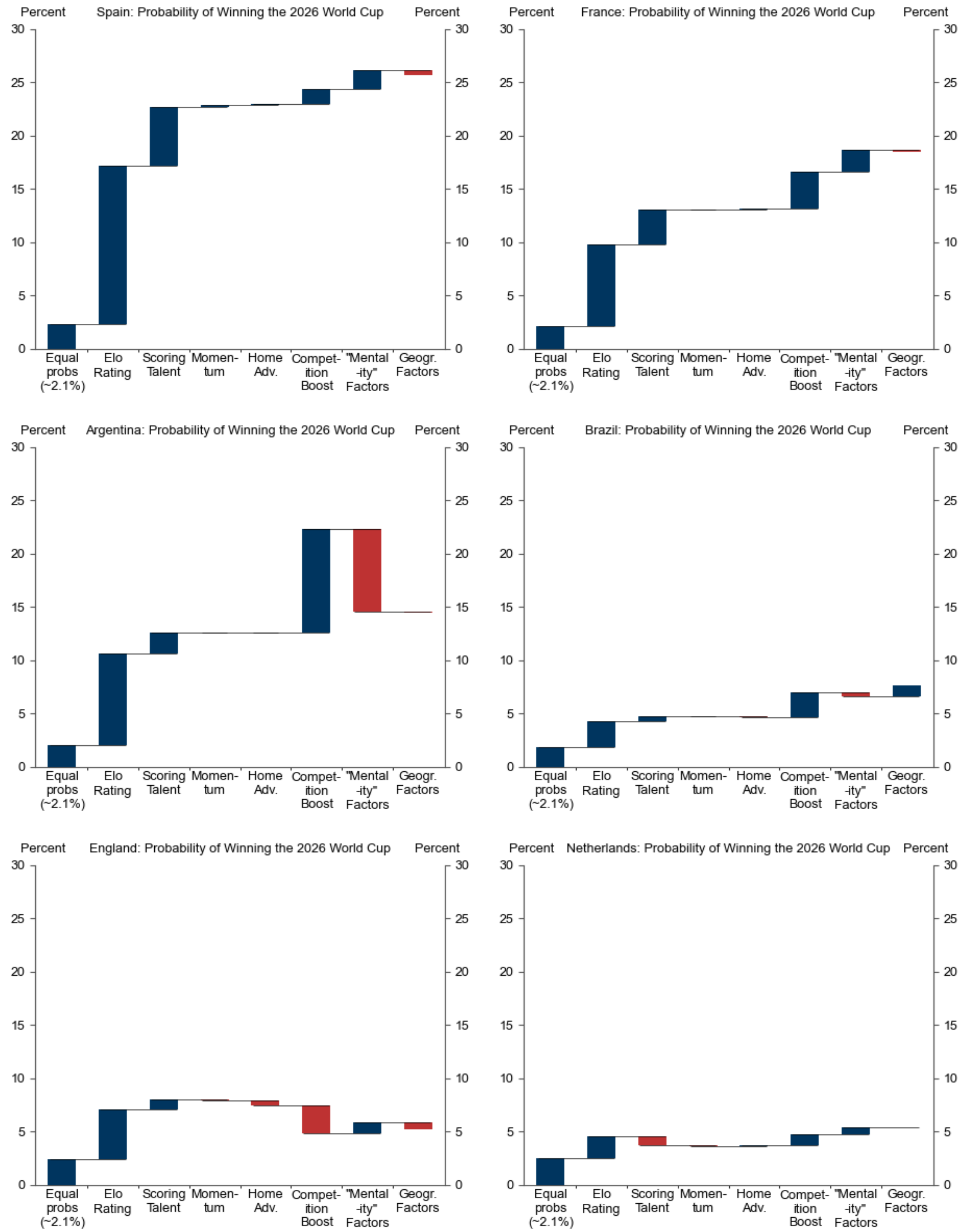
Model-Implied Probabilities of Advancing in the 2026 World Cup (as of 29 May 2026)													
Team	R32	R16	QF	SF	Final	Winner	Team	R32	R16	QF	SF	Final	Winner
Spain	99.1	80.6	65.1	55.3	37.5	25.7	Uzbekistan	54.5	18.9	6.2	2.0	0.6	0.2
France	96.2	82.4	61.9	47.0	28.9	18.9	Australia	59.4	28.1	9.9	2.6	0.8	0.2
Argentina	96.5	70.0	57.3	42.4	27.1	14.3	Czechia	68.7	32.5	10.0	2.3	0.7	0.2
Brazil	96.1	66.9	45.3	28.4	15.9	7.6	Scotland	64.3	18.6	7.2	2.0	0.6	0.1
Netherlands	94.3	61.6	45.9	22.7	11.0	5.2	Algeria	50.3	13.7	5.0	1.5	0.4	0.1
England	94.1	65.9	37.7	22.2	11.5	5.0	Iran	71.6	30.3	9.7	2.2	0.5	0.1
Portugal	90.2	61.4	37.9	21.0	11.0	4.8	Panama	58.0	17.9	5.3	1.6	0.4	0.1
Germany	95.0	67.0	32.1	19.9	9.5	4.5	Jordan	47.9	13.7	5.2	1.7	0.4	0.1
Colombia	87.4	51.5	26.7	13.2	5.9	2.2	South Korea	71.2	34.5	10.6	2.0	0.5	0.1
Croatia	87.4	49.1	22.5	11.3	4.9	1.7	Egypt	60.7	22.0	6.2	1.4	0.3	0.1
Norway	78.0	47.4	23.7	11.8	4.8	1.6	Sweden	54.2	13.0	5.0	1.4	0.4	0.1
Mexico	95.6	68.0	33.3	10.2	3.4	0.8	Ivory Coast	54.8	14.9	3.3	0.9	0.2	0.0
Senegal	70.4	37.5	16.8	7.3	2.6	0.8	DR Congo	34.9	8.7	2.1	0.6	0.1	0.0
Ecuador	84.6	42.1	15.3	6.8	2.3	0.8	Bosnia and Herz.	63.2	22.7	6.2	1.1	0.2	0.0
Belgium	86.3	51.6	23.9	6.8	2.3	0.7	Cape Verde	42.2	7.9	2.3	0.6	0.1	0.0
Switzerland	88.9	54.2	21.8	6.7	2.2	0.7	New Zealand	52.9	17.2	4.3	0.8	0.2	0.0
Turkiye	77.3	45.9	21.8	6.8	2.2	0.6	Tunisia	39.0	7.4	2.3	0.6	0.1	0.0
United States	69.5	38.9	17.1	5.4	1.7	0.5	Haiti	29.4	5.4	1.3	0.3	0.1	0.0
Japan	78.9	30.7	16.0	5.6	1.8	0.5	Iraq	22.0	5.7	1.4	0.3	0.1	0.0
Austria	69.4	23.9	11.2	4.7	1.6	0.4	Saudi Arabia	35.0	5.5	1.4	0.2	0.0	0.0
Uruguay	79.9	26.5	12.9	5.3	1.7	0.4	South Africa	31.8	8.7	1.6	0.2	0.0	0.0
Paraguay	67.0	35.4	13.9	4.1	1.2	0.3	Curacao	31.5	6.1	1.0	0.2	0.0	0.0
Morocco	76.1	26.9	11.9	3.7	1.1	0.3	Ghana	26.4	5.1	1.0	0.2	0.0	0.0
Canada	90.7	50.4	18.4	4.7	1.2	0.3	Qatar	27.5	5.5	0.9	0.1	0.0	0.0

Source: Goldman Sachs Global Investment Research

Exhibit 8 provides more insight into the results by breaking down the probabilities of winning for the top four teams in a "waterfall chart" format. Our key predictions are:

- Spain is supported by its top Elo rating (52 points more than Argentina and 84 points above France) and scoring talent.
- Argentina is supported by a strong competition boost and the second-highest Elo rating but affected negatively by the winner's slump, as reigning World Cup winners consistently underperform (the only recent exception is France 2018-22).
- France boasts elite scoring talent but suffers from likely facing Spain earlier than Argentina (already in the semi-finals).
- England faces the twin obstacles of historical underperformance at World Cups and the likelihood of facing Mexico in hot and high-altitude Mexico City, which results in a net-drag from geographical factors on the Three Lions' chances.
- Germany is the only top team with no contribution from offensive talent. This is because relative to others they have fewer big scorers.

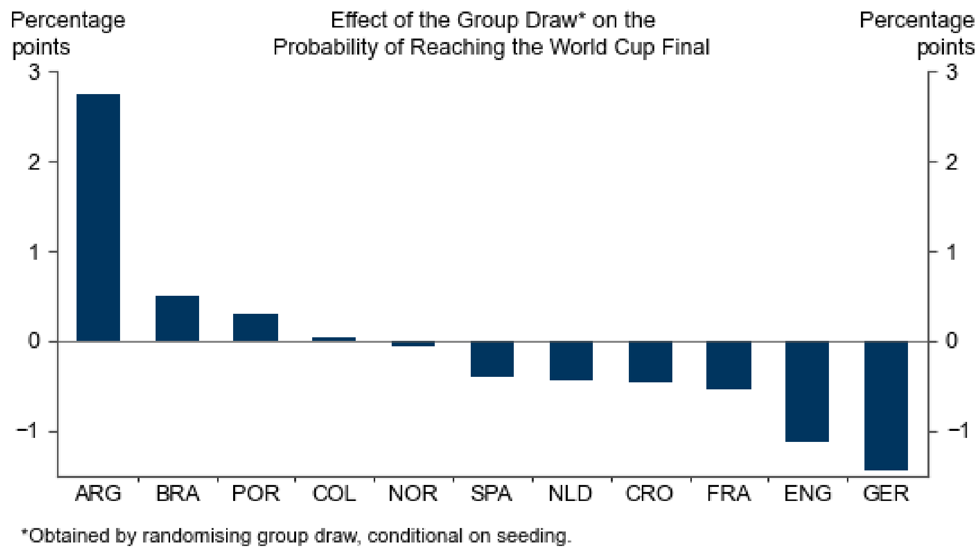
Exhibit 8: Breaking Down the Winning Probability



Source: elorating.net, Goldman Sachs Global Investment Research

An important (and highly debated) factor in all major football tournaments is the draw. We leverage our model to see how the draw affects the probabilities of reaching the final by re-running simulations in a scenario where groups are randomised (conditional on seeding, Exhibit 9). Our analysis suggests that Germany was the least lucky team in the draw as they are predicted to face France in the round of 16. Argentina was, according to our model, the luckiest as they are likely to face top-ranked Spain only in the final.

Exhibit 9: The Luck of the Draw



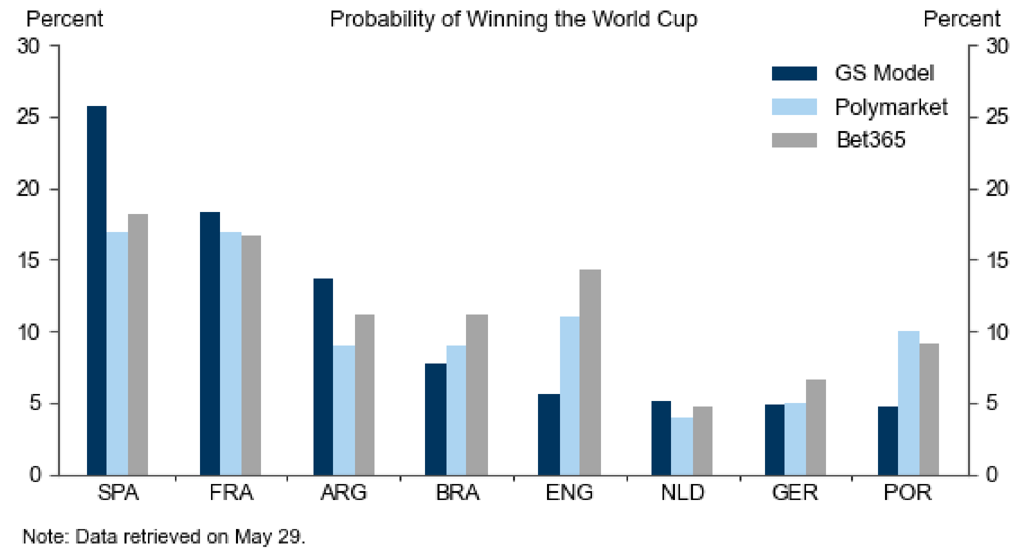
We randomise group composition, conditional on having in all groups one team from each pot.

Source: Goldman Sachs Global Investment Research

We can also compare our probabilities to the market (Exhibit 10). We are overweight Spain and Argentina and underweight England and Portugal.

While our expanded model accounts for a larger set of factors than our previous attempts, it remains largely blind to the non-attacking talent (*e.g. the depth of the French and Portuguese midfield and wings or the role of good keepers in a penalty shootout*), health (*will Lamine Yamal's recent injury weigh on his performance in the World Cup?*), individual momentum (*will the many French players at PSG benefit from their deep run in the Champions League? Will Messi and Ronaldo be competitive after a few years away from top European football?*), or managerial experience (*an Ancelotti effect for Brazil?*). Furthermore, our model treats the number of goals scored by each team involved in a game as independent distributions, which leads to several predicted draws.

Exhibit 10: Comparing Our Model Probabilities to the Market

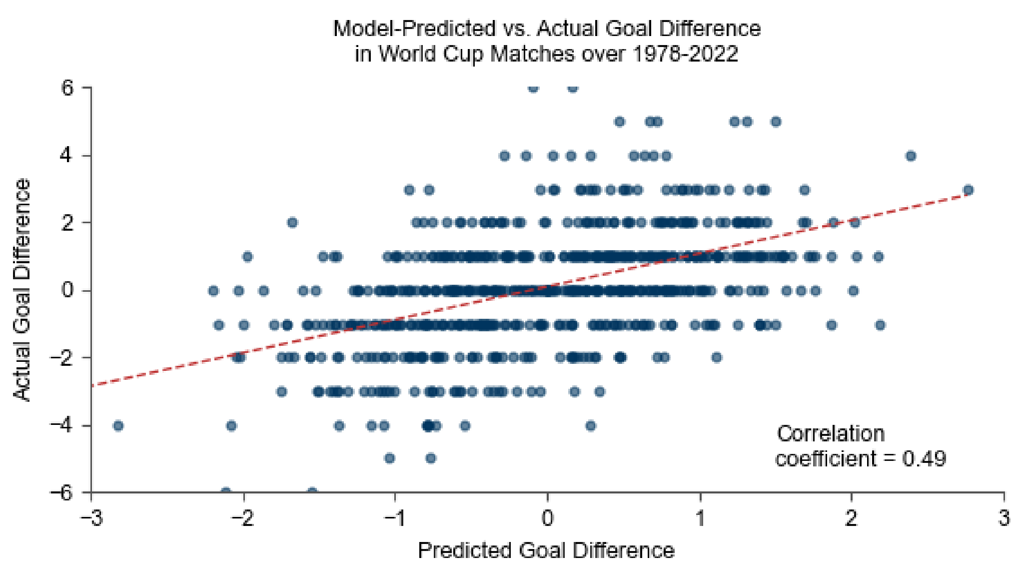


Source: Goldman Sachs Global Investment Research, Polymarket, Bet365

Another useful check is to evaluate what our model would have suggested ahead of the 2022 World Cup. The predicted winning probabilities ahead of the start of the tournament would have been Brazil (24% chances of winning), followed by Argentina (21%) and France (19%). This broadly mirrored the bookmakers' implied odds of Brazil (18%), Argentina (13%), and France (12%).

We can also look at the correlation between actual and predicted goal difference historically, which is ultimately what our model is trained to predict. The correlation of our model is 49% across all World Cup games since 1978, 43% in 2018, and 45% in 2022. While the correlation remains modest—given football's inherent unpredictability—the model improved over previous frameworks (Exhibit 11).

Exhibit 11: Assessing Our Model's Past Performance



Source: Goldman Sachs Global Investment Research

Throughout the tournament, in a short comment after each match day, we will be commenting on how the incoming results compare to our predictions, preview the games of the following day, and update the most-likely and probabilistic predictions of our model.

Jan Hatzius

Giovanni Pierdomenico

Sven Jari Stehn

Appendix: The Group Stage Standings

We leverage official FIFA rules to tie breaks and rank the 3rd-placed teams, with the top 8 advancing to the round of 32. In the event our model is unable to distinguish between teams using FIFA rules (e.g. in a group with predicted draws in all games on a rounded basis, given we don't model red and yellow cards), we rely on the predicted unrounded goal difference.

Exhibit 12: The Predicted Group Standings

FIFA World Cup 2026 - Final Group Standings (Prediction as of 29 May 2026)

Qualified (1st / 2nd) | Qualified as best-3rd | Eliminated

Group A

Pos	Team	GF	GA	GD	Pts	Q
1	Mexico	6	2	+4	9	Top-2
2	South Korea	3	4	-1	2	Top-2
3	Czechia	3	4	-1	2	Best-3rd
4	South Africa	2	4	-2	2	Out

Group B

Pos	Team	GF	GA	GD	Pts	Q
1	Canada	5	3	+2	7	Top-2
2	Switzerland	4	2	+2	5	Top-2
3	Bosnia and H.	3	4	-1	2	Best-3rd
4	Qatar	2	5	-3	1	Out

Group C

Pos	Team	GF	GA	GD	Pts	Q
1	Brazil	6	3	+3	9	Top-2
2	Morocco	3	4	-1	2	Top-2
3	Scotland	3	4	-1	2	Best-3rd
4	Haiti	3	4	-1	2	Out

Group D

Pos	Team	GF	GA	GD	Pts	Q
1	Turkiye	3	3	+0	3	Top-2
2	USA	3	3	+0	3	Top-2
3	Paraguay	3	3	+0	3	Best-3rd
4	Australia	3	3	+0	3	Out

Group E

Pos	Team	GF	GA	GD	Pts	Q
1	Germany	6	3	+3	9	Top-2
2	Ecuador	4	4	+0	4	Top-2
3	Ivory Coast	3	4	-1	2	Out
4	Curacao	3	5	-2	1	Out

Group F

Pos	Team	GF	GA	GD	Pts	Q
1	Netherlands	5	3	+2	7	Top-2
2	Japan	3	3	+0	3	Top-2
3	Sweden	3	4	-1	2	Best-3rd
4	Tunisia	3	4	-1	2	Out

Group G

Pos	Team	GF	GA	GD	Pts	Q
1	Belgium	3	3	+0	3	Top-2
2	Iran	3	3	+0	3	Top-2
3	Egypt	3	3	+0	3	Best-3rd
4	New Zealand	3	3	+0	3	Out

Group H

Pos	Team	GF	GA	GD	Pts	Q
1	Spain	8	0	+8	9	Top-2
2	Uruguay	2	4	-2	2	Top-2
3	Cape Verde	2	5	-3	2	Out
4	Saudi Arabia	2	5	-3	2	Out

Group I

Pos	Team	GF	GA	GD	Pts	Q
1	France	7	2	+5	9	Top-2
2	Norway	4	4	+0	4	Top-2
3	Senegal	4	4	+0	4	Best-3rd
4	Iraq	2	7	-5	0	Out

Group J

Pos	Team	GF	GA	GD	Pts	Q
1	Argentina	6	3	+3	9	Top-2
2	Austria	3	4	-1	2	Top-2
3	Algeria	3	4	-1	2	Out
4	Jordan	3	4	-1	2	Out

Group K

Pos	Team	GF	GA	GD	Pts	Q
1	Portugal	5	3	+2	7	Top-2
2	Colombia	4	3	+1	5	Top-2
3	Uzbekistan	3	4	-1	2	Best-3rd
4	DR Congo	3	5	-2	1	Out

Group L

Pos	Team	GF	GA	GD	Pts	Q
1	England	5	3	+2	7	Top-2
2	Croatia	5	3	+2	7	Top-2
3	Panama	3	5	-2	1	Out
4	Ghana	3	5	-2	1	Out

Source: Goldman Sachs Global Investment Research

Disclosure Appendix

Reg AC

We, Jan Hatzius, Giovanni Pierdomenico and Sven Jari Stehn, hereby certify that all of the views expressed in this report accurately reflect our personal views, which have not been influenced by considerations of the firm's business or client relationships.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

Contributing Authors: Jan Hatzius Goldman Sachs & Co. LLC, Giovanni Pierdomenico Goldman Sachs International, Sven Jari Stehn Goldman Sachs International.

Unless otherwise stated, the individuals listed in the Contributing Authors disclosure of this report are analysts in Goldman Sachs' Global Investment Research division.

Disclosures

Regulatory disclosures

Disclosures required by United States laws and regulations

See company-specific regulatory disclosures above for any of the following disclosures required as to companies referred to in this report: manager or co-manager in a pending transaction; 1% or other ownership; compensation for certain services; types of client relationships; managed/co-managed public offerings in prior periods; directorships; for equity securities, market making and/or specialist role. Goldman Sachs trades or may trade as a principal in debt securities (or in related derivatives) of issuers discussed in this report.

The following are additional required disclosures: **Ownership and material conflicts of interest:** Goldman Sachs policy prohibits its analysts, professionals reporting to analysts and members of their households from owning securities of any company in the analyst's area of coverage. **Analyst compensation:** Analysts are paid in part based on the profitability of Goldman Sachs, which includes investment banking revenues. **Analyst as officer or director:** Goldman Sachs policy generally prohibits its analysts, persons reporting to analysts or members of their households from serving as an officer, director or advisor of any company in the analyst's area of coverage. **Non-U.S. Analysts:** Non-U.S. analysts may not be associated persons of Goldman Sachs & Co. LLC and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with a subject company, public appearances and trading in securities covered by the analysts.

Additional disclosures required under the laws and regulations of jurisdictions other than the United States

The following disclosures are those required by the jurisdiction indicated, except to the extent already made above pursuant to United States laws and regulations. **Australia:** Goldman Sachs Australia Pty Ltd and its affiliates are not authorised deposit-taking institutions (as that term is defined in the Banking Act 1959 (Cth)) in Australia and do not provide banking services, nor carry on a banking business, in Australia. This research, and any access to it, is intended only for "wholesale clients" within the meaning of the Australian Corporations Act, unless otherwise agreed by Goldman Sachs. In producing research reports, members of Global Investment Research of Goldman Sachs Australia may attend site visits and other meetings hosted by the companies and other entities which are the subject of its research reports. In some instances the costs of such site visits or meetings may be met in part or in whole by the issuers concerned if Goldman Sachs Australia considers it is appropriate and reasonable in the specific circumstances relating to the site visit or meeting. To the extent that the contents of this document contains any financial product advice, it is general advice only and has been prepared by Goldman Sachs without taking into account a client's objectives, financial situation or needs. A client should, before acting on any such advice, consider the appropriateness of the advice having regard to the client's own objectives, financial situation and needs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests and a copy of Goldman Sachs' Australian Sell-Side Research Independence Policy Statement are available at: https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html. **Brazil:** Disclosure information in relation to CVM Resolution n. 20 is available at https://www.gs.com/worldwide/brazil/area/gir/index.html. Where applicable, the Brazil-registered analyst primarily responsible for the content of this research report, as defined in Article 20 of CVM Resolution n. 20, is the first author named at the beginning of this report, unless indicated otherwise at the end of the text. **Canada:** This information is being provided to you for information purposes only and is not, and under no circumstances should be construed as, an advertisement, offering or solicitation by Goldman Sachs & Co. LLC for purchasers of securities in Canada to trade in any Canadian security. Goldman Sachs & Co. LLC is not registered as a dealer in any jurisdiction in Canada under applicable Canadian securities laws and generally is not permitted to trade in Canadian securities and may be prohibited from selling certain securities and products in certain jurisdictions in Canada. If you wish to trade in any Canadian securities or other products in Canada please contact Goldman Sachs Canada Inc., an affiliate of The Goldman Sachs Group Inc., or another registered Canadian dealer. **Hong Kong:** Further information on the securities of covered companies referred to in this research may be obtained on request from Goldman Sachs (Asia) L.L.C. **India:** Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (India) Securities Private Limited, Research Analyst - SEBI Registration Number INH000001493, 10th Floor, Ascent-Worli, Sudam Kalu Ahire Marg, Worli, Mumbai-400 025, India, Corporate Identity Number U74140MH2006FTC160634, Phone +91 22 6616 9000, Fax +91 22 6616 9001. Goldman Sachs may beneficially own 1% or more of the securities (as such term is defined in clause 2 (h) the Indian Securities Contracts (Regulation) Act, 1956) of the subject company or companies referred to in this research report. Investment in securities market are subject to market risks. Read all the related documents carefully before investing. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Goldman Sachs (India) Securities Private Limited compliance officer and investor grievance contact details can be found at: https://www.goldmansachs.com/worldwide/india/documents/Grievance-Redressal-and-Escalation-Matrix.pdf, and a copy of the annual audit compliance report can be found at this link: https://publishing.gs.com/content/site/india-annual-compliance-report.html. **Japan:** See below. **Korea:** This research, and any access to it, is intended only for "professional investors" within the meaning of the Financial Services and Capital Markets Act, unless otherwise agreed by Goldman Sachs. Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (Asia) L.L.C., Seoul Branch. **New Zealand:** Goldman Sachs New Zealand Limited and its affiliates are neither "registered banks" nor "deposit takers" (as defined in the Reserve Bank of New Zealand Act 1989) in New Zealand. This research, and any access to it, is intended for "wholesale clients" (as defined in the Financial Advisers Act 2008) unless otherwise agreed by Goldman Sachs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests is available at: https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html. **Russia:** Research reports distributed in the Russian Federation are not advertising as defined in the Russian legislation, but are information and analysis not having product promotion as their main purpose and do not provide appraisal within the meaning of the Russian legislation on appraisal activity. Research reports do not constitute a personalized investment recommendation as defined in Russian laws and regulations, are not addressed to a specific client, and are prepared without analyzing the financial circumstances, investment profiles or risk profiles of clients. Goldman Sachs assumes no responsibility for any investment decisions that may be taken by a client or any other person based on this research report. **Singapore:** Goldman Sachs (Singapore) Pte. (Company Number: 198602165W), which is regulated by the Monetary Authority of Singapore, accepts legal responsibility for this research, and should be contacted with respect to any matters arising from, or in connection with, this research. **Taiwan:** This material is for reference only and must not be reprinted without permission. Investors should carefully consider their own investment risk. Investment results are the responsibility of the individual investor. **United Kingdom:** Persons who would be categorized as retail clients in the United Kingdom, as such term is defined in the rules of the Financial Conduct Authority, should read this research in conjunction with prior Goldman Sachs research on the covered companies referred to herein and should refer to the risk warnings that have been sent to them by Goldman Sachs International. A copy of these risks warnings, and a glossary of certain financial terms used in this report, are available from Goldman Sachs International on request.

European Union and United Kingdom: Disclosure information in relation to Article 6 (2) of the European Commission Delegated Regulation (EU) (2016/958) supplementing Regulation (EU) No 596/2014 of the European Parliament and of the Council (including as that Delegated Regulation is implemented into United Kingdom domestic law and regulation following the United Kingdom's departure from the European Union and the European Economic Area) with regard to regulatory technical standards for the technical arrangements for objective presentation of investment recommendations or other information recommending or suggesting an investment strategy and for disclosure of particular interests or indications of conflicts of interest is available at https://www.gs.com/disclosures/europeanpolicy.html which states the European Policy for Managing Conflicts of Interest in Connection with Investment Research.

Japan: Goldman Sachs Japan Co., Ltd. is a Financial Instrument Dealer registered with the Kanto Financial Bureau under registration number Kinsho 69, and a member of Japan Securities Dealers Association, Financial Futures Association of Japan Type II Financial Instruments Firms Association, and Investment Management Association of Japan. Sales and purchase of equities are subject to commission pre-determined with clients plus consumption tax. See company-specific disclosures as to any applicable disclosures required by Japanese stock exchanges, the Japanese Securities Dealers Association or the Japanese Securities Finance Company.

Global product; distributing entities

Goldman Sachs Global Investment Research produces and distributes research products for clients of Goldman Sachs on a global basis. Analysts based in Goldman Sachs offices around the world produce research on industries and companies, and research on macroeconomics, currencies, commodities and portfolio strategy. This research is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Brazil by Goldman Sachs do Brasil Corretora de Títulos e Valores Mobiliários S.A.; Public Communication Channel Goldman Sachs Brazil: 0800 727 5764 and / or contatogoldmanbrasil@gs.com. Available Weekdays (except holidays), from 9am to 6pm. Canal de Comunicação com o Público Goldman Sachs Brasil: 0800 727 5764 e/ou contatogoldmanbrasil@gs.com. Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h; in Canada by Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs (Asia) L.L.C.; in India by Goldman Sachs (India) Securities Private Ltd.; in Japan by Goldman Sachs Japan Co., Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C., Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in Russia by OOO Goldman Sachs; in Singapore by Goldman Sachs (Singapore) Pte. (Company Number: 198602165W); and in the United States of America by Goldman Sachs & Co. LLC. Goldman Sachs International has approved this research in connection with its distribution in the United Kingdom.

Goldman Sachs International ("GSI"), authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA, has approved this research in connection with its distribution in the United Kingdom.

European Economic Area: Goldman Sachs Bank Europe SE ("GSBE") is a credit institution incorporated in Germany and, within the Single Supervisory Mechanism, subject to direct prudential supervision by the European Central Bank and in other respects supervised by German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and Deutsche Bundesbank and disseminates research within the European Economic Area.

General disclosures

This research is for our clients only. Other than disclosures relating to Goldman Sachs, this research is based on current public information that we consider reliable, but we do not represent it is accurate or complete, and it should not be relied on as such. The information, opinions, estimates and forecasts contained herein are as of the date hereof and are subject to change without prior notification. We seek to update our research as appropriate, but various regulations may prevent us from doing so. Other than certain industry reports published on a periodic basis, the large majority of reports are published at irregular intervals as appropriate in the analyst's judgment.

Goldman Sachs conducts a global full-service, integrated investment banking, investment management, and brokerage business. We have investment banking and other business relationships with a substantial percentage of the companies covered by Global Investment Research. Goldman Sachs & Co. LLC, the United States broker dealer, is a member of SIPC (https://www.sipc.org).

Our salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients and principal trading desks that reflect opinions that are contrary to the opinions expressed in this research. Our asset management area, principal trading desks and investing businesses may make investment decisions that are inconsistent with the recommendations or views expressed in this research.

We and our affiliates, officers, directors, and employees will from time to time have long or short positions in, act as principal in, and buy or sell, the securities or derivatives, if any, referred to in this research, unless otherwise prohibited by regulation or Goldman Sachs policy.

The views attributed to third party presenters at Goldman Sachs arranged conferences, including individuals from other parts of Goldman Sachs, do not necessarily reflect those of Global Investment Research and are not an official view of Goldman Sachs.

Any third party referenced herein, including any salespeople, traders and other professionals or members of their household, may have positions in the products mentioned that are inconsistent with the views expressed by analysts named in this report.

This research is focused on investment themes across markets, industries and sectors. It does not attempt to distinguish between the prospects or performance of, or provide analysis of, individual companies within any industry or sector we describe.

Any trading recommendation in this research relating to an equity or credit security or securities within an industry or sector is reflective of the investment theme being discussed and is not a recommendation of any such security in isolation.

This research is not an offer to sell or the solicitation of an offer to buy any security in any jurisdiction where such an offer or solicitation would be illegal. It does not constitute a personal recommendation or take into account the particular investment objectives, financial situations, or needs of individual clients. Clients should consider whether any advice or recommendation in this research is suitable for their particular circumstances and, if appropriate, seek professional advice, including tax advice. The price and value of investments referred to in this research and the income from them may fluctuate. Past performance is not a guide to future performance, future returns are not guaranteed, and a loss of original capital may occur. Fluctuations in exchange rates could have adverse effects on the value or price of, or income derived from, certain investments.

Certain transactions, including those involving futures, options, and other derivatives, give rise to substantial risk and are not suitable for all investors. Investors should review current options and futures disclosure documents which are available from Goldman Sachs sales representatives or at https://www.theocc.com/about/publications/character-risks.jsp and https://www.goldmansachs.com/disclosures/cftc_fcm_disclosures. Transaction costs may be significant in option strategies calling for multiple purchase and sales of options such as spreads. Supporting documentation will be supplied upon request.

Differing Levels of Service provided by Global Investment Research: The level and types of services provided to you by Goldman Sachs Global Investment Research may vary as compared to that provided to internal and other external clients of GS, depending on various factors including your individual preferences as to the frequency and manner of receiving communication, your risk profile and investment focus and perspective (e.g., marketwide, sector specific, long term, short term), the size and scope of your overall client relationship with GS, and legal and regulatory constraints. As an example, certain clients may request to receive notifications when research on specific securities is published, and certain clients may request that specific data underlying analysts' fundamental analysis available on our internal client websites be delivered to them electronically through data feeds or otherwise. No change to an analyst's fundamental research views (e.g., ratings, price targets, or material changes to earnings estimates for equity securities), will be communicated to any client prior to inclusion of such information in a research report broadly disseminated through electronic publication to our internal client websites or through other means, as necessary, to all clients who are entitled to receive such reports.

All research reports are disseminated and available to all clients simultaneously through electronic publication to our internal client websites. Not all research content is redistributed to our clients or available to third-party aggregators, nor is Goldman Sachs responsible for the redistribution of our research by third party aggregators. For research, models or other data related to one or more securities, markets or asset classes (including related services) that may be available to you, please contact your GS representative or go to https://research.gs.com.

Disclosure information is also available at https://www.gs.com/research/hedge.html or from Research Compliance, 200 West Street, New York, NY 10282.